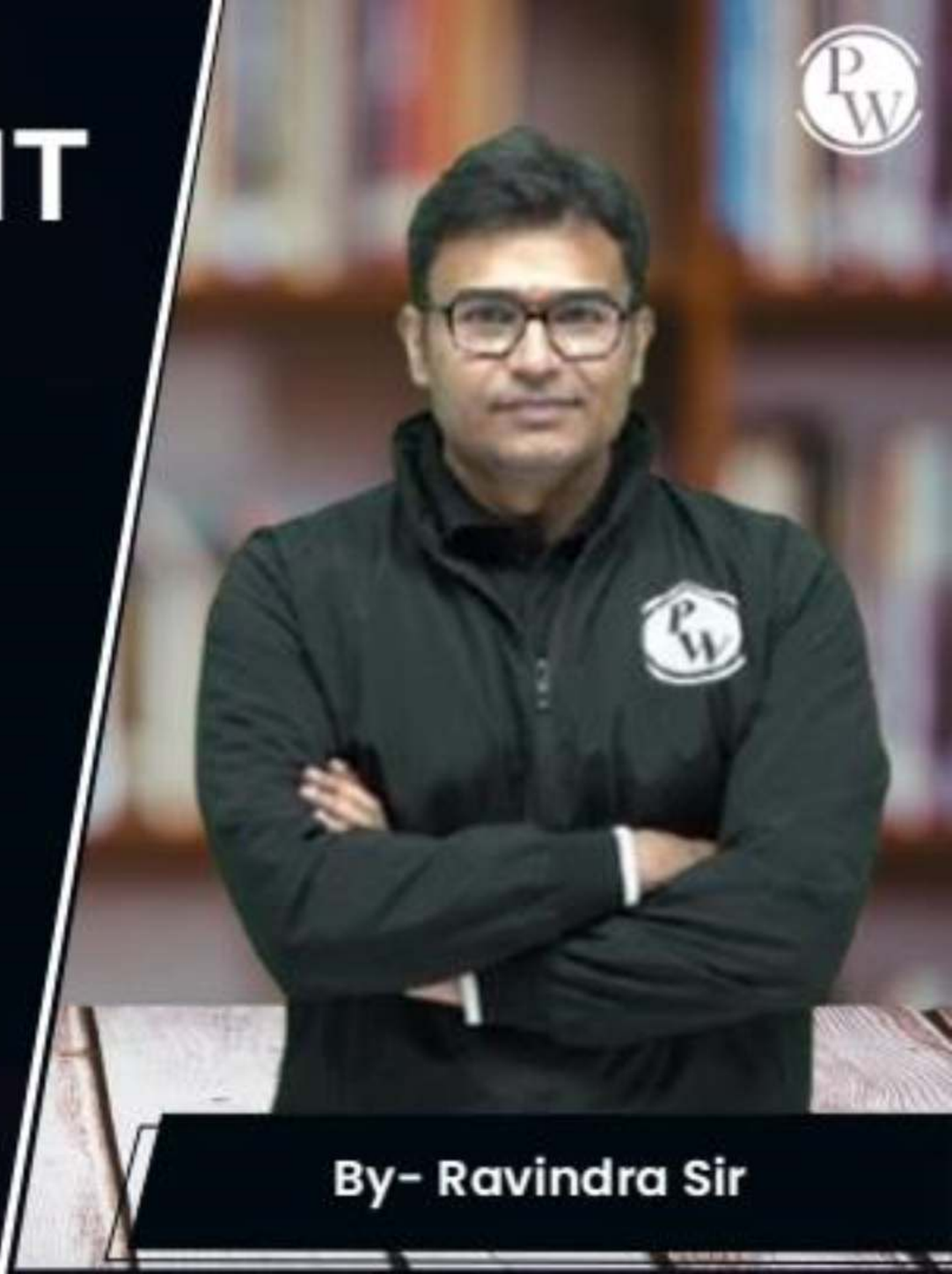
Computer Science & IT

COMPUTER NETWORKS (CN)

Flow Control Methods

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Flow Control

Topics to be Covered



- Topic
- Topic
- Topic

Flow Control

Join Telegram

@RBRSIRPW





Inspiring Stories : Baby Halder

Baby Halder – The Maid Who Became a Bestselling Author

Background: Child bride at 12. Mother of 3 by 20. Worked as a domestic help in Delhi to survive.

Struggles: Abusive marriage, abandoned with children. Cleaned houses during the day, raised kids, wrote at night with discarded notebooks.

Achievements: Wrote "A Life Less Ordinary", a raw memoir translated into over 20 languages. Invited to global lit fests.

Impact: She didn't have peace, privilege, or polish — just persistence. If she could write her way out of despair, you can code your way out of confusion.



Inspiring Stories : Pratish Datta

Pratish Datta (IIT Kharagpur – M.Sc. Mathematics)

Background: Visually impaired—lost eyesight by college; depended on his mother for learning early on.

Education: M.Sc in Mathematics from IIT Kharagpur.

Career Achievements: Topped his cohort with a GPA of 9.87; won Professor Jagadish Chandra Bose Memorial Gold Medal.

Impact: Demonstrated exceptional intellectual resilience; turned his disability into an intellectual strength.



Inspiring Stories : Riddhima Paul

Riddhima Paul (IIT Kanpur – BS in Mathematics & Scientific Computing)

Background: Born with spinal muscular atrophy (SMA), wheelchair-bound from childhood; faced school denial.
Education: Joined IIT Kanpur after a preparatory year at IIT Indore.
Career Achievements: Scored exceptionally well in Class 10 and 12; qualified for IIT despite missing the direct rank by one mark.
Impact: A living example of academic excellence and resilience overcoming physical challenges and social bias.



Inspiring Stories : Peter Norman

Peter Norman – The White Man Who Stood Still

Background: Australian sprinter at the 1968 Mexico Olympics, a time when civil rights tensions were boiling.

Struggles: Chose to silently stand with Black athletes Tommie Smith and John Carlos during their iconic Black Power salute — wore the same badge.

Achievements: Though he won Silver, he was ostracized by Australia, left out of future Olympics, and forgotten for decades.

Impact: He never raised his fist — he just stood still with conviction. If he could hold his ground under global pressure, you can hold your line under exam stress.

Stop and wait

+

To timer (ARQ)

+

Seq → data

+

Seq → Ack

$$\eta = \frac{1}{1+2a}$$

$$= \frac{1}{1+2T_p/T_t}$$

$$= \frac{1}{1+2\left(\frac{d}{v}\times\frac{B}{L}\right)}$$

$$Th = \eta * BW$$

or

$$BW\ util = \eta * BW$$

or

$$eff\ BW = \eta * BW$$

S&W → Lans

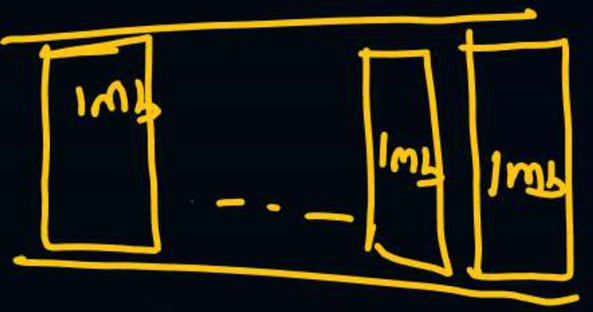
L → Busty data → effi

Capacity of a channel / wire / link :-

BW = 1bps Tp = 10 sec



BW = 1mbps Tp = 10 secs



Capacity :- The maximum of bits that can be present in any channel is called capacity of the channel

Channel → medium through which data passes

Capacity = BW * Tp → half duplex

= 2 * BW * Tp → full duplex

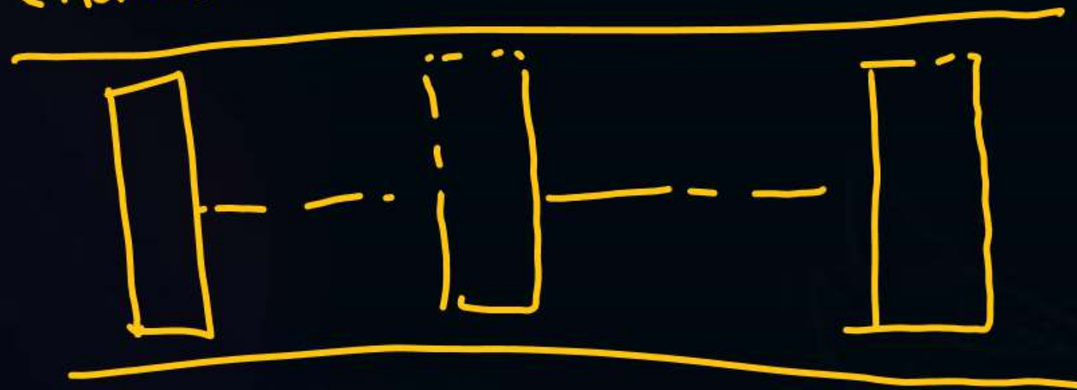
full duplex = $2 * BW * T_p$

Cap ↑ → Thick channel

Cap ↓ → Thin channel

BW ↑ Cap ↑

T_p ↑ Cap ↑



full duplex
→ data can go in both directions at the same time

Half duplex :-
Data can go in only one direction at a time

→ If channel is thick (Cap ↑), then efficiency of S&W is low

$$\eta = \frac{1}{1+2a} = \frac{1}{1+2*T_p/T_t}$$

$$= \frac{1}{1+\left(2*\frac{T_p*B}{L}\right)}$$

$\frac{T_p*B}{L}$ → Cap of channel

Cap ↑ → η ↓

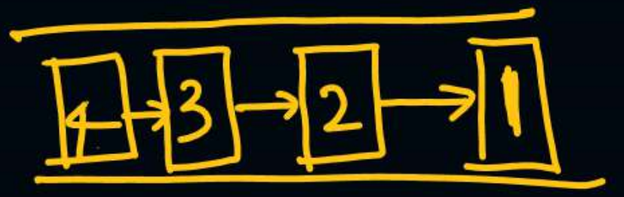
Cap ↓ → η ↑



To increase efficiency we use pipeling

Stop and wait





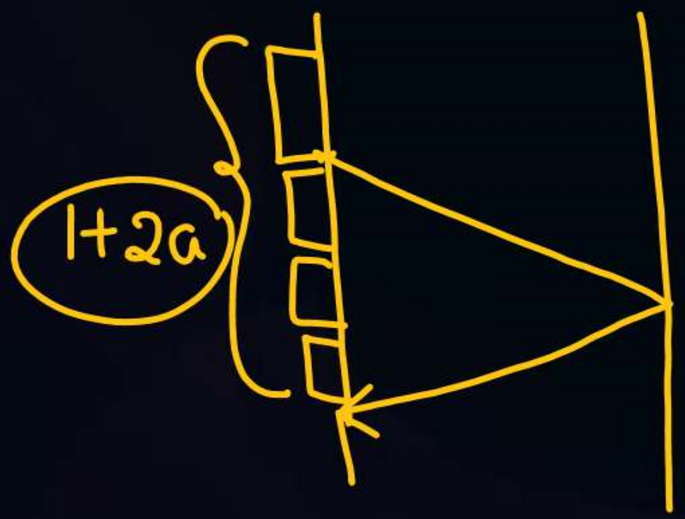
1 packet transmit $\rightarrow T_t$ sec

1 sec $\rightarrow \frac{1}{T_t}$ packets

$T_t + 2 * T_p$ sec $\rightarrow \frac{T_t + 2 * T_p}{T_t}$

$1 + 2 \frac{T_p}{T_t}$

$1 + 2a$



In stop and wait

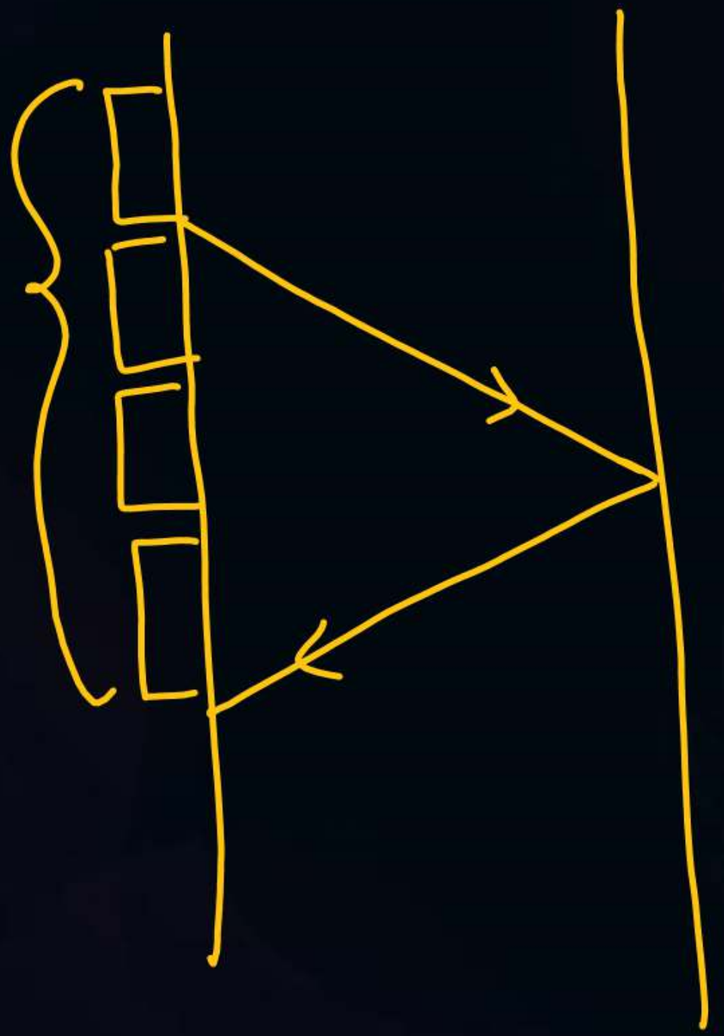
In one cycle — (1+2a) packets

$$\eta = \frac{1}{1+2a}$$

If we have to increase efficiency, then

Send more packets

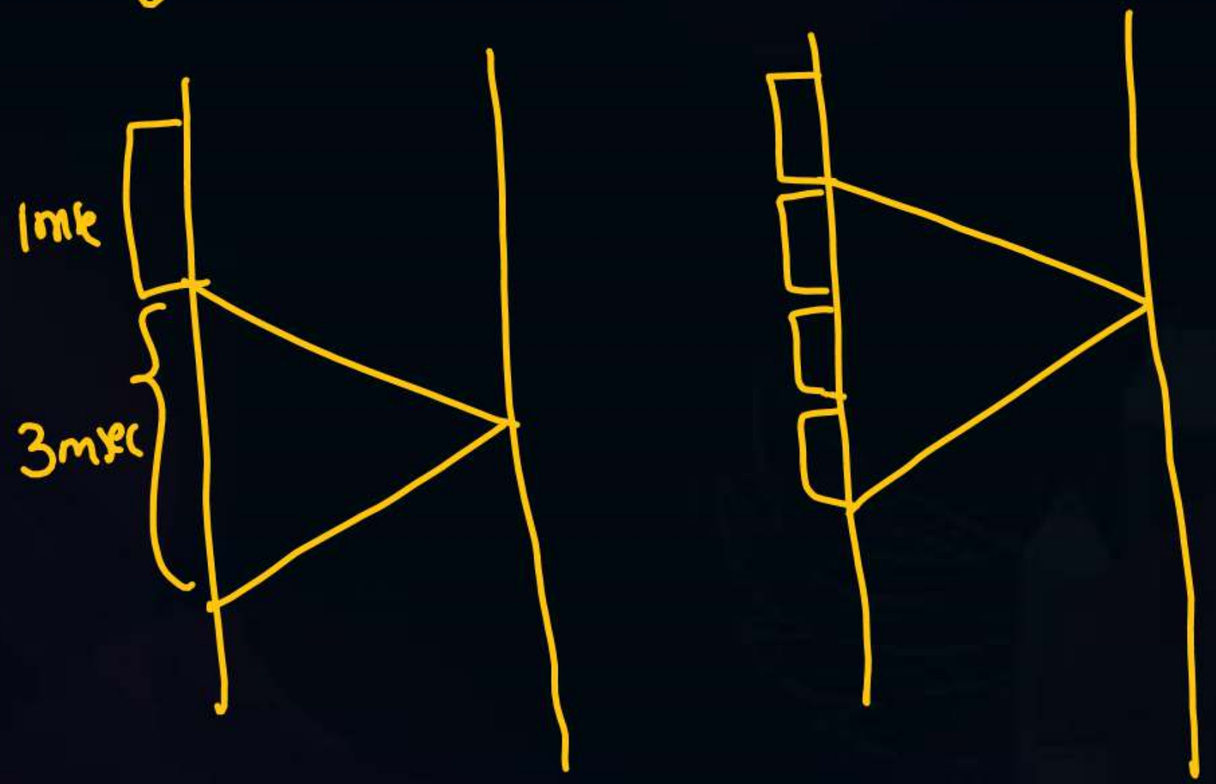
This is called pipeling

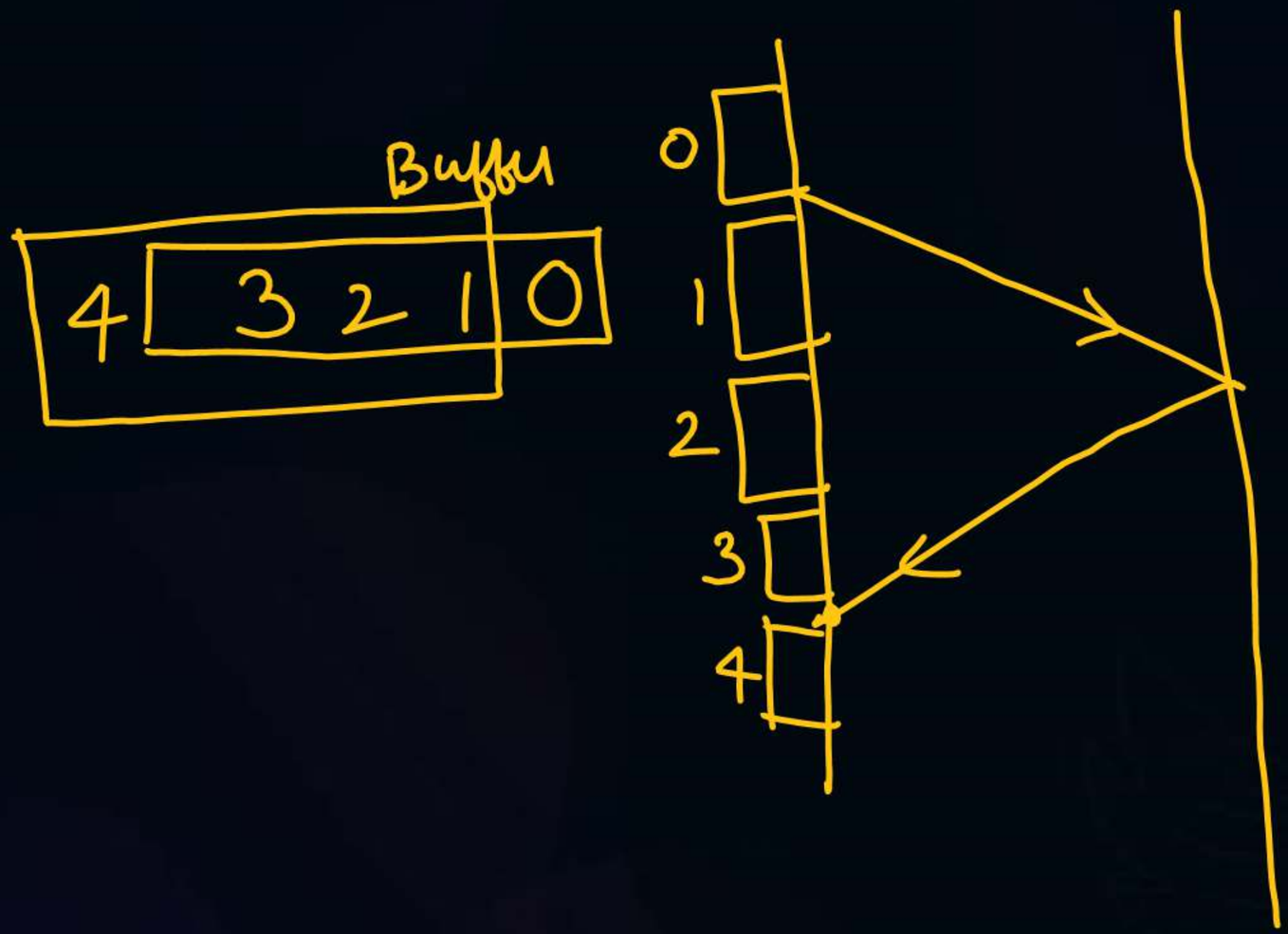


If we use pipelining and sent many packets then eff $\eta = 100\%$.

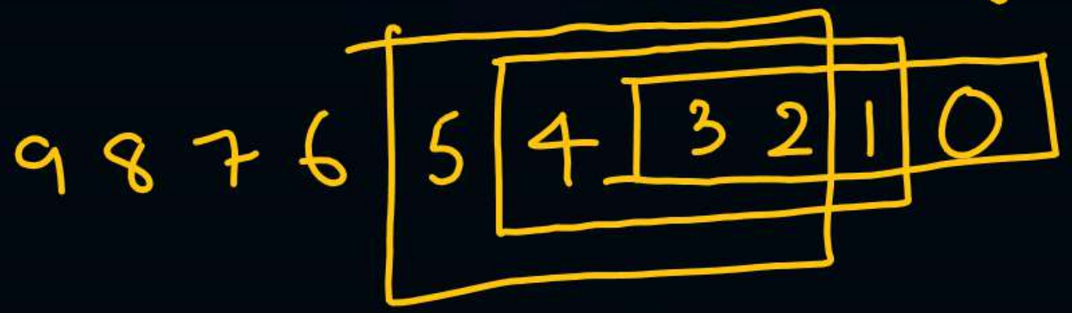
$T_t = 1\,ms \quad T_p = 1.5\,ms = 3/2\,ms$

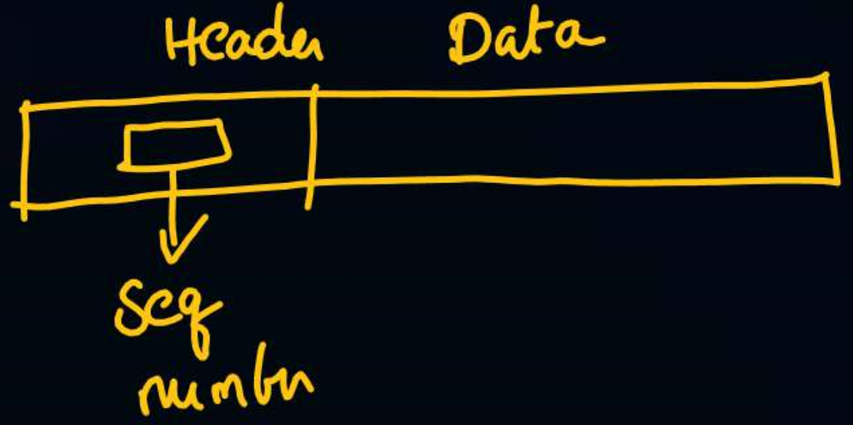
$$\eta = \frac{1}{1+2a} = \frac{1}{1+2*T_p/T_t} = \frac{1}{1+2*3/2} = \frac{1}{4}$$

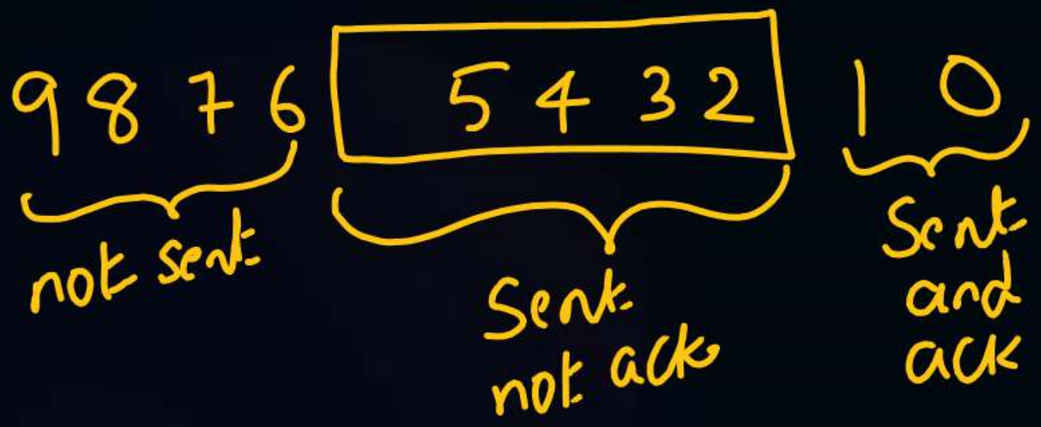



Buffer
4
3
2
1
0
0
1
2
3
4

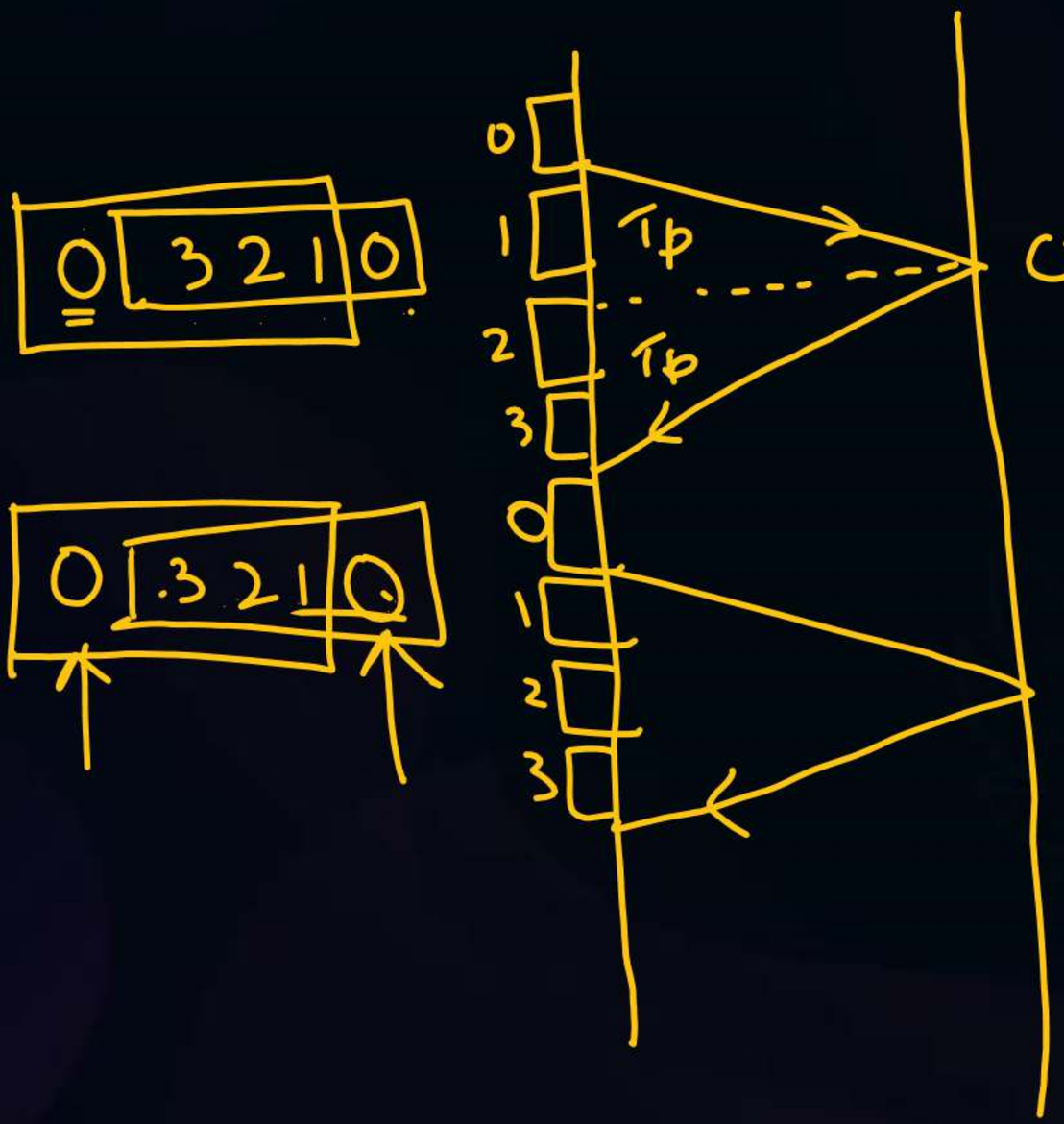
Sliding window protocol







$T_t = 1\,ms$ $T_p = 1.5\,ms$ $2 * T_p = 3\,ms$



4 - Seq numbs

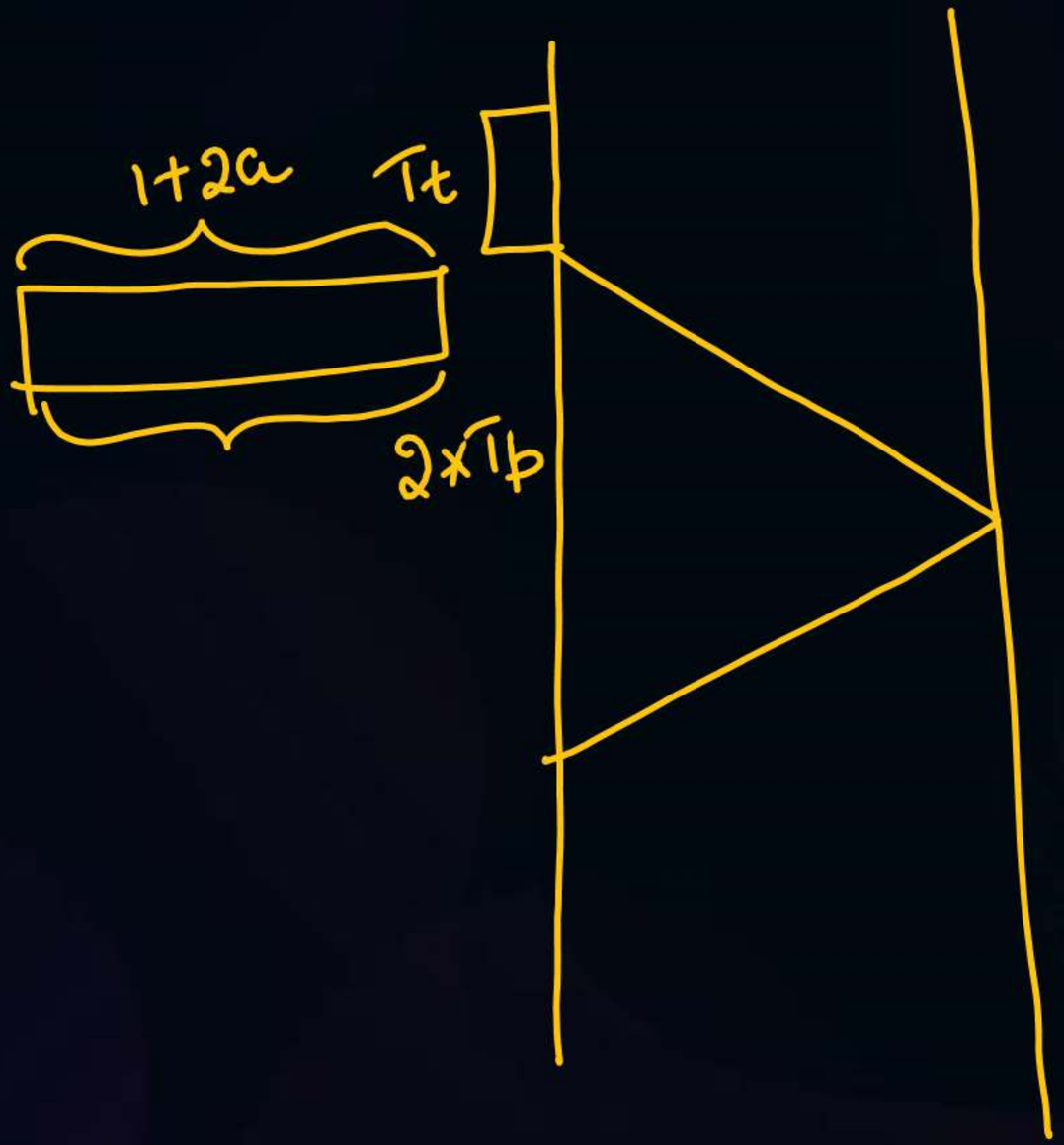
bits - 2 bits

00 - 0

01 - 1

10 - 2

11 - 3



$T_t + 2 * T_p \rightarrow (1+2a)$ packets

Seq number = $1+2a$

Bits for seq num = $\lceil \log_2(1+2a) \rceil$

$T_t = 1ms \quad T_p = 49.5\ msec$



100% η

→ 100 seq numbers

100ms → 100 packets

Seq − 100 ✓

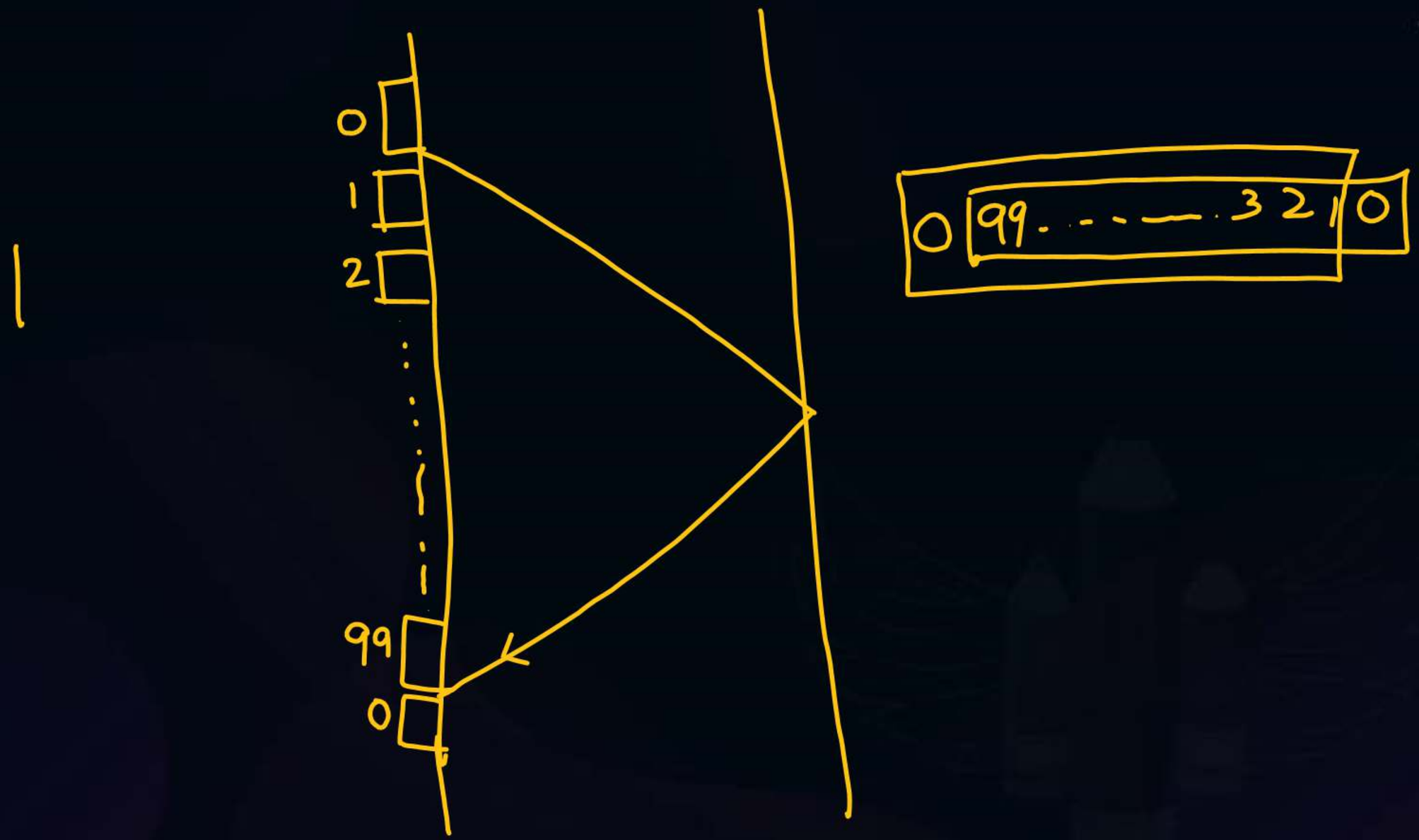
Bits in seq numbr fields

$\lceil \log_2 100 \rceil$

= 7 bits − 128

Window = 1 + 2a = 1 + 2 * 49.5

= 100 ✓


0
1
2
99
0
0 99 . . . . 3 2 1 0

$\eta = ?$ $T_t = 1ms$, $T_p = 49.5ms$,

no of bits in seq number field = 6 bits

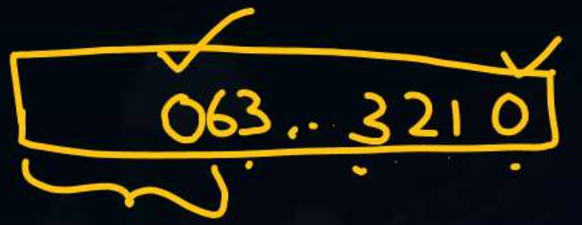
N → no of bits in seq number field

Window Size = $1 + 2a = 100$

$2^6 = 64$ seq numbers

$$\eta = \frac{\min(1+2a, 2^N)}{1+2a}$$





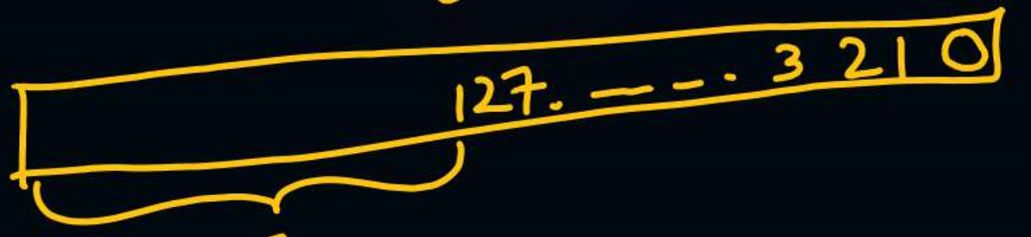
$$\eta = \frac{64}{100}$$

$T_t = 1ms \quad T_p = 99.5\,ms \quad \underline{\underline{N = 7}}$, $\eta = ?$

$$W_S = 1 + 2a = 1 + 2 * \frac{T_p}{T_t} = 1 + 2 * 99.5 = 200$$

Th of effective BW of BW utilized





$Seq = 2^7 = 128$

$\frac{128}{200} = 64\%$

$$\eta = \frac{\min(1+2a, 2^N)}{1+2a} = \frac{128}{200} = 64\%$$

$BW = 4mbps \quad \underline{Th} = \eta * BW = 0.64 \times 4mbps$

Sliding window protocol :-

Sliding window protocol is a theoritical concept

Practically it is implemented as GBN & SR

G B N

↓ | |

go back N

S R

| |

Selective Repeat

Reject


GBN
1
2
3
SR
1
2
3

GBN:-

RBR Rule 1) Sender window size is 'N'

$W_S > 1$ ✓ $W_S = 1$ (GBN)

GB10 $W_S = 10$

$T_t = 1ms$ $T_p = 49.5ms$, η of GB10

$$\eta = \frac{10}{1+2a} = \frac{10}{100} = 10\%$$

mbps ↑ MBps ↑

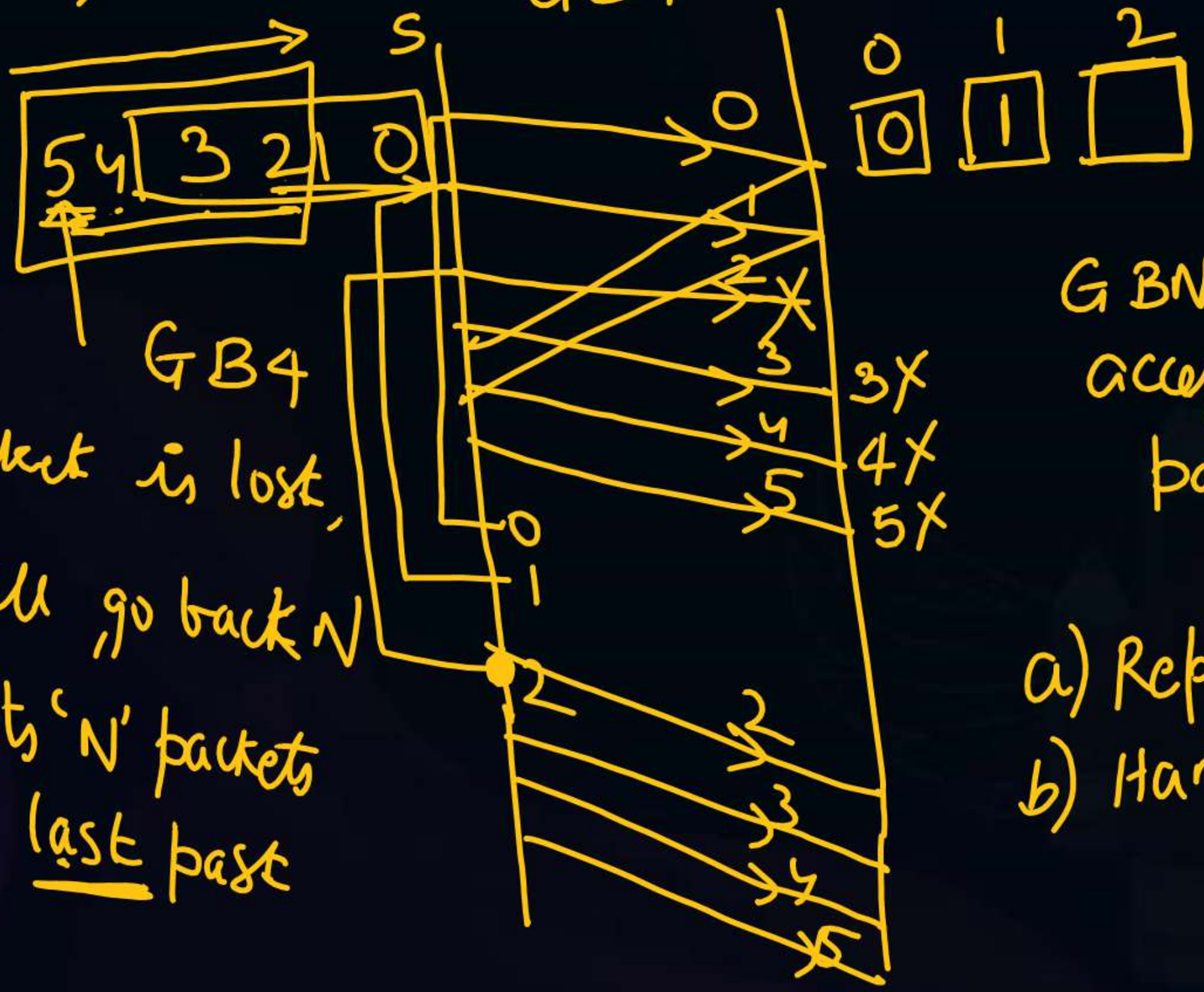
BW = 40 mbps

$$Th = \eta \times BW = 0.1 \times 40 mbps = 4 mbps$$

GBN:-

RBR Rule 2) Rec window Size is always 1

GB4

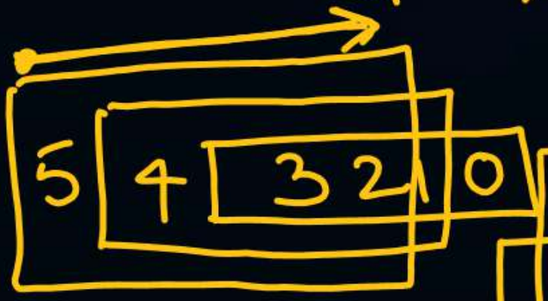


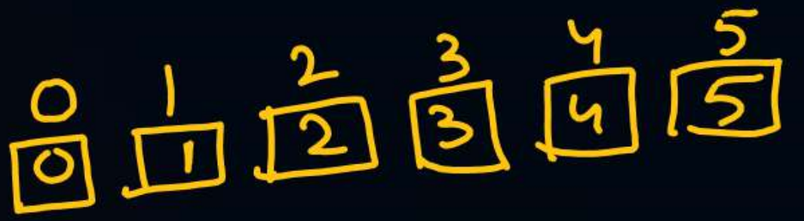
If a packet is lost, Sender will go back N Retransmits 'N' packets Starting last past

GBN, Rec will never accept out of order packets

a) Repeat
b) Handsome
c) Repeat and handsome

GB4 RBR Rule 2)

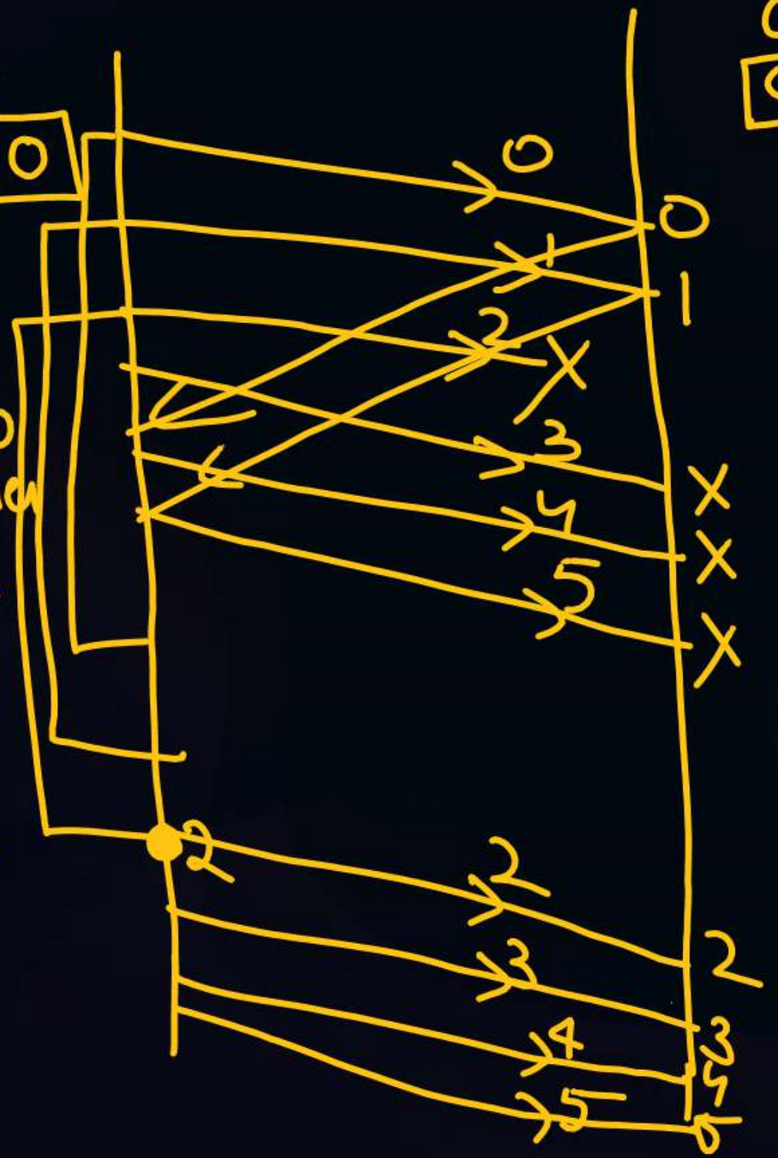




GBN, Rec will never ACC out of order packets

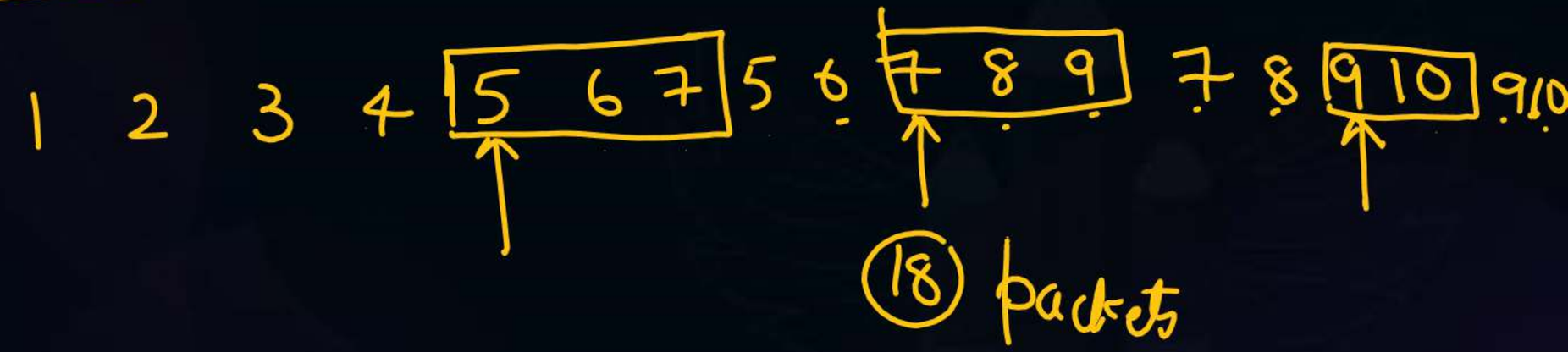
GB4 TO Timer

GBN, if a packet is lost, entire window will be retransmitted

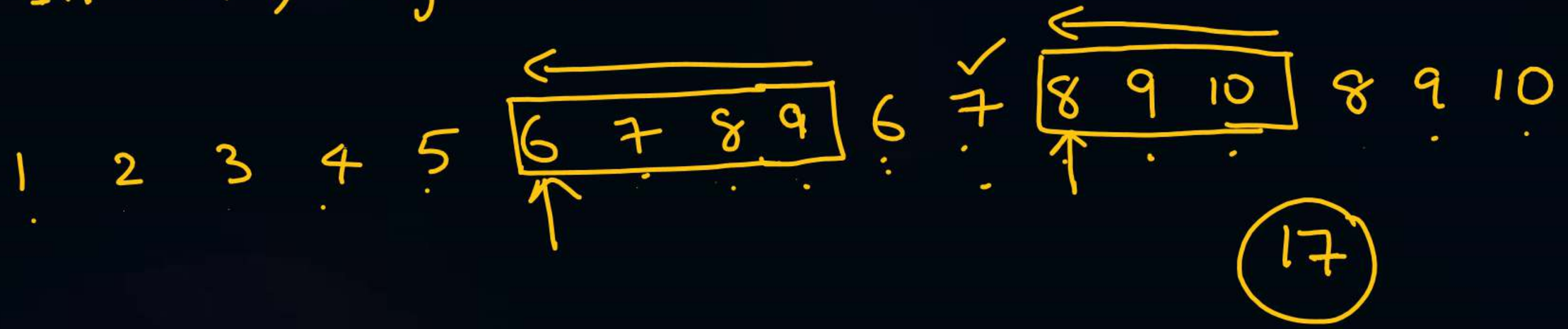


In GB3, if every 5th packet that is transmitted is lost and if we send 10 packets, then total how many packets are transmitted?





In GB4, every 6th packet is lost. → 10 packets



THANK - YOU